Course outline

About NPTEL

How does an NPTEL online course work?

Week 0

Week 1

Week 2

Week 3

Week 4

- Bias - I
- Bias - II
- Source of Bias
- Bias-Handson
- Week 4- Lecture Materials
- Quiz: Week 4: Assignment 4
- Week 4 Feedback Form: Responsible & Safe AI Systems
- Assignment 4 solution

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Download videos

Week 4: Assignment 4

The due date for submitting this assignment has passed.

Due on 2024-08-

As per our records you have not submitted this assignment.

1) Which of the following define bias? (Select all that apply)

- [ ] Systematic favoritism towards certain groups
- [ ] Random errors in data collection
- [ ] Neutral and unbiased judgment
- [ ] Systematic exclusion of certain data points
- [ ] Random selection of samples
- [ ] Systematic deviation from rationality in judgment

No, the answer is incorrect.
Score: 0
Accepted Answers:
Systematic favoritism towards certain groups
Systematic deviation from rationality in judgment

2) Which of the following is one of the ways to measure bias

- ( ) Random sampling
- ( ) Cross-Validation
- ( ) Using Benchmark datasets
- ( ) Chi-Square test

No, the answer is incorrect.
Score: 0
Accepted Answers:
Using Benchmark datasets

3) Which of the following statements is true?

- ( ) Demographics do not influence the perception of bias.
- ( ) Demographics can influence the perception of bias.
- ( ) Perception of bias is unaffected by demographics.
- ( ) Demographics and bias perception are unrelated.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Demographics can influence the perception of bias.

4) What does each pair in the CrowS-Pairs dataset consist of?

- ( ) One stereotype sentence and one neutral sentence
- ( ) One biased sentence and one unbiased sentence
- ( ) One stereotype sentence and one opposite stereotype sentence
- ( ) One stereotype sentence and one less stereotype sentence

No, the answer is incorrect.
Score: 0
Accepted Answers:
One stereotype sentence and one less stereotype sentence

5) The statement “Women are bad drivers” is a

- ( ) Stereotype
- ( ) Anti-Stereotype
- ( ) Non-Stereotype
- ( ) Neutral Statement

No, the answer is incorrect.
Score: 0
Accepted Answers:
Stereotype

6) Which of the following is NOT a common source of bias in data?

- ( ) Historical inequities reflected in the data
- ( ) Data collection methods that over-represent certain groups
- ( ) Balanced representation of all demographic groups

Labelling errors or subjective annotations

No, the answer is incorrect.
Score: 0

Accepted Answers:
Balanced representation of all demographic groups

7) At which stage of the machine learning process can bias be introduced by over- or under-representing certain groups?

- Data Collection
- Data Pre-processing
- Data Annotation
- All of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
All of the above

8) Which of the following characteristics can make a statement biased?

- Being stereotypical
- Being socially aligned
- Having inaccurate information
- Having emotional language

No, the answer is incorrect.
Score: 0

Accepted Answers:
Being stereotypical

9) One of the reasons bias exists in models is:

- Algorithm complexity
- Data
- Model architecture
- Training duration

No, the answer is incorrect.
Score: 0

Accepted Answers:
Data

10) When is bias a problem?

- When it reduces the models performance
- When it has negative impact
- When it aligns with train data but not test data
- When it increases the complexity

No, the answer is incorrect.
Score: 0

Accepted Answers:
When it has negative impact

11) Which of the following is the correct setting for contrastive learning?

- [ ] Irrespective of the sentences, minimise the distance between their embeddings
- [ ] Irrespective of the sentences, maximise the distance between their embeddings
- [ ] If sentences are similar, minimise the distance between their embeddings
- [ ] If sentences are different, minimise the distance between their embeddings
- [ ] If sentences are different, maximise the distance between their embeddings
- [ ] If sentences are different, minimise the distance between their embeddings

No, the answer is incorrect.
Score: 0

Accepted Answers:
If sentences are similar, minimise the distance between their embeddings
If sentences are different, maximise the distance between their embeddings